

ChE 422 – Biochemical Engineering –Spring 2026

Student Learning Outcome 3: An ability to communicate effectively with a range of audiences (written communication)

Rubric assessment of literature review

Summary of Results:

Rubric Assessment Table

Performance Indicator (PI)	Level 3: Satisfactory	Level 2: Developing	Level 1: Unsatisfactory
1. Articulation of ideas	Articulates ideas clearly and concisely	Articulates ideas, but writing is somewhat disjointed, superfluous or difficult to follow	Text rambles, points made are only understood with repeated reading, and key points are not organized
2. The organization of the written materials	Organizes written materials in a logical sequence to enhance the reader's comprehension (paragraphs, subheading, etc.)	Material are generally organized well, but paragraphs combine multiple thoughts or sections and sub-sections are not identified clearly	Little or no structure or organization; no subheadings or proper paragraph structure used
3. The Use of graphs, tables, and diagrams	Uses graphs, tables, and diagrams to support points-to explain, interpret, and assess information	Uses graphs, tables, and diagrams, but only in a few instances are they applied to support, explain or interpret information	Graphs, tables or diagrams are used, but no reference is made to them
4. The presentation of the written work	Written work is presented neatly and professionally	Work is not neatly presented throughout	Work is not presented neatly
5. Grammar and spelling	Grammar and spelling are correct	One or two spelling/grammar errors per page	Spelling/grammar errors present throughout more than 1/3 of the paper
6. Writing style	Uses good professional writing style	Style is informal or inappropriate, jargon is used, improper voice, tense...	The writing style is inappropriate for the audience and for the assignment

7. Prescribed format	Conforms to the prescribed format	The prescribed format is only followed in some portions of the paper	The prescribed format is not followed
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RESULTS BY PERFORMANCE INDICATOR:Final Lab Report							
Rubric item (PI)	1	2	3	4	5	6	7
Mapped SLO	3	3	3	3	3	3	3
Average	2.83	2.83	2.17	3.00	3.00	3.00	2.90
Std Dev	0.00	0.00	1.78	2.00	2.00	2.00	1.73

RESULTS FOR STUDENT OUTCOME 3: ALL ASSESSMENTS	
Average Score	2.82
Standard dev.	0.93

Assignment Description: Written Review

Students are required to complete a written review based on one of the following selected topics:

1. Gene therapies
2. Synthetic biology
3. Nanomedicine
4. AI-guided biochemical processes

The review should provide a critical analysis of current developments, challenges, and future directions within the chosen area.

Detailed guidelines regarding the structure, formatting template, and assessment rubric were provided to students in advance.